Mason Tran

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Summary

Cyber Operations and Security Operations Center (SOC) analyst with hands-on experience in log analysis, network traffic inspection, and incident investigation across endpoint and network telemetry. Experienced in SIEM-based detection, ELK stack pipelines, and scripting for threat analysis and alert triage. Strong understanding of the attack lifecycle, security monitoring, and incident scoping within enterprise environments.

Education

University of Arizona, Tucson, AZ **Expected 2026**

CAST Center of Academic Excellence – Cyber Operations Bachelor's degree (in progress) – Cyber Operations / Cybersecurity

Experience

Cybersecurity Analyst – Security Operations Center (SOC) **Apr 2025 – Present**

University of Arizona *Tucson, AZ*

- Analyze real-time security events across endpoint and network telemetry using ELK-based SIEM workflows, supporting detection and investigation of suspicious activity.
- Perform log analysis on network, system, and protocol-level data (Zeek, Suricata-style telemetry) to scope incidents, identify attacker behavior, and assess impact.
- Investigate anomalous traffic patterns, lateral movement indicators, and unauthorized activity using packet captures and structured log data.
- Develop and maintain parsing pipelines and normalization logic (ECS-style schemas) to improve detection accuracy and analyst efficiency.
- Write Python and Bash scripts to automate alert enrichment, data correlation, and investigative workflows.
- Identify benign activity patterns and reduce false positives by refining detection logic and filters.
- Experience analyzing endpoint and network detection telemetry analogous to EDR/NDR platforms through host logs, packet inspection, and behavioral indicators.
- Performed basic static analysis of suspicious artifacts (hashing, strings, metadata) during investigative workflows.

Certified Repair Apple Technician **Apr 2024 – Apr 2025**

University of Arizona *Tucson, AZ*

- Performed diagnostics, failure analysis, component-level troubleshooting, and thermal performance checks on logic boards, displays, voltage circuits, SSD controllers, and power delivery paths.
- Conducted system diagnostics and failure analysis at the OS and firmware level, reinforcing understanding of endpoint behavior, persistence mechanisms, and system integrity.
- Demonstrated knowledge of circuit behavior, current flow, voltage regulation, and component reliability
- Supported endpoint integrity and reliability by validating system behavior after repair, reinforcing incident response concepts related to host trust and post-compromise recovery.
- Interpreted schematics, board diagrams, and electrical test points to validate board repairs and ensure adherence to OEM specifications.

  Led IT process workflows for software re-images, firmware resets, device boot diagnostics, and post-repair validation testing.
- Successfully passed Apple certification exams and internal repair qualification tests, demonstrating proficiency in technical assembly and process operations consistent with defense-grade manufacturing standards.

Assembly Technician / Electro Mechanical Assembler **Oct 2022 – Nov 2023**

Integrated Magnetics *Tucson, AZ*

- Performed precision electro-mechanical assembly following detailed technical documentation and quality standards.
- Conducted inspections and documented results to support traceability and compliance.
- Worked in a high-security, process-driven environment requiring strict adherence to procedures.

Projects

Wilma AI | Node.js, Express, Fireflies API, OpenAI API, PDFKit, Git **HackAZ, March 2025**

- Developed an intelligent assistant for a challenge on improving the UofA that automatically joins Zoom lectures, transcribes discussions, and delivers polished study notes as PDFs. Built a full-stack Node.js/Express app integrating the Fireflies API for real-time meeting transcription, OpenAI for summarization and key concept extraction, and custom PDF generation for clean note formatting.
- Designed CLI tools and backend workflows to schedule recordings, process transcripts, and email notes to users, streamlining lecture review for students. Project submitted for Hack Arizona 2025, focusing on enhancing the University of Arizona student experience.

React Portfolio | React, HTML, SCSS, JavaScript **Present**

- Developed a responsive portfolio website using Create React App with a modular, component-based architecture.
- Integrated Email.js API to facilitate direct user communication.
- Enhanced user engagement with dynamic interactions using React hooks and state management.

SQL Web Checker | Python, Tkinter **Present**

- Created a Python-based web vulnerability scanner with a Tkinter GUI to test for SQL injection, XSS, and command injection.
- Implemented multi-threading for responsiveness and regex-based error detection for precise vulnerability assessment.

Password Encrypter Manager | *P*ython Tkinter **Present**

- Developed a secure password manager utilizing AES encryption with PBKDF2 key derivation and PKCS7 padding.
- Designed an intuitive Tkinter-based GUI featuring secure login, auto-login, and CRUD operations for account management.

Technical Skills

Security & Incident Response:

- **SIEM-based alert triage and investigation**
- **Network traffic analysis (PCAP, protocol inspection)**
- **Log correlation and incident scoping**
- **Endpoint and network behavioral analysis**
- **Attack lifecycle (initial access, lateral movement, impact)**
- **Detection tuning and false-positive reduction**

Languages:

Python, Bash, C, C++, Java, JavaScript, Assembly

Tools & Platforms:

ELK Stack, Linux, Git, Zeek, Suricata-style telemetry

Frameworks & Dev:

React, Node.js, JUnit